



ApuShape User Guide

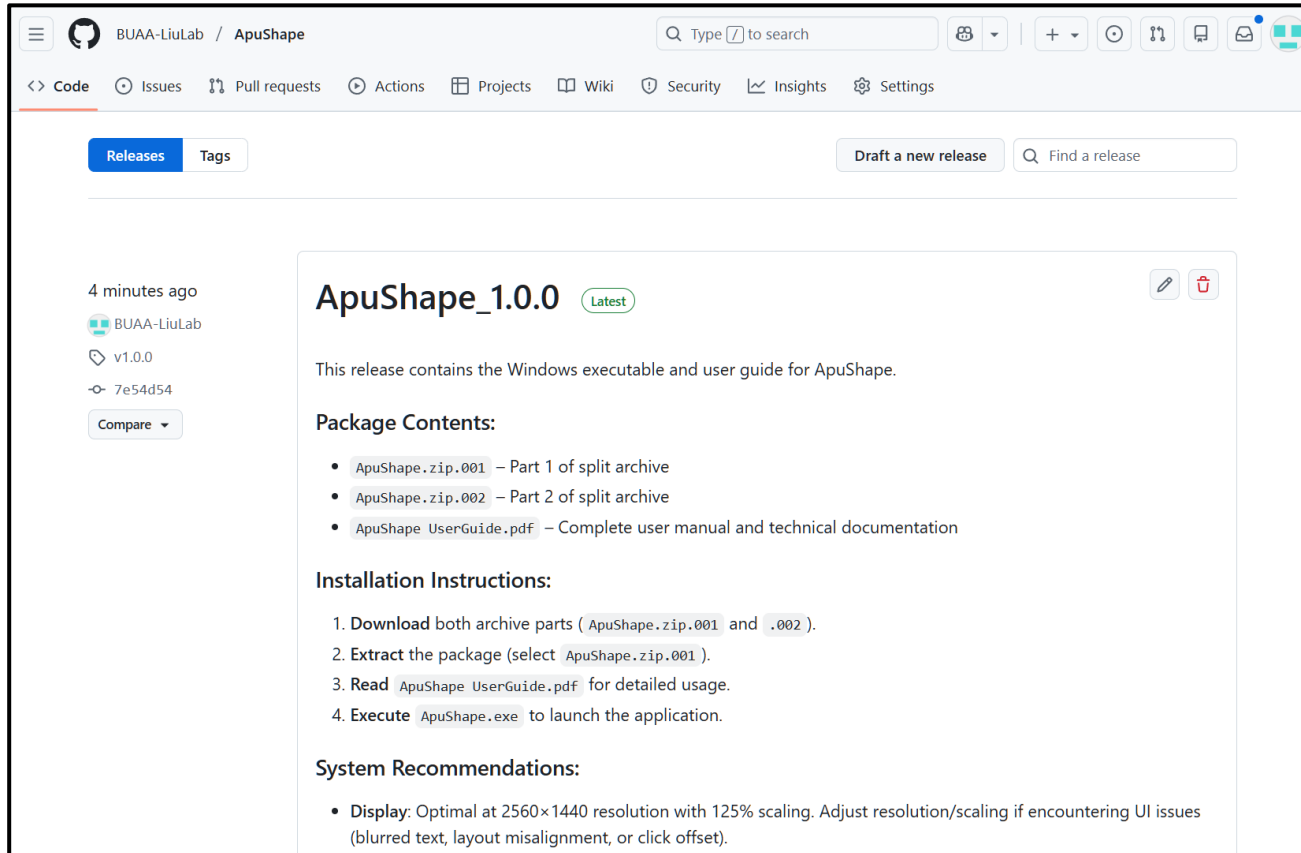
Version 1.0.0

2025.11

Contents

- **Software Download**
- **Main interface of ApuShape**
- **How to run ApuShape**
 - ◇ **Manually annotate datasets**
 - ◇ **Annotate datasets with ShapeStar**
 - ◇ **Correct annotations and retrain model with ShapeActive**
 - ◇ **Refine annotations with ShapeRefine**

Software Download



Download the software from the ApuShape releases page: <https://github.com/BUAA-LiuLab/ApuShape/releases>. Select the latest version, download and extract the package, then run **ApuShape.exe**.

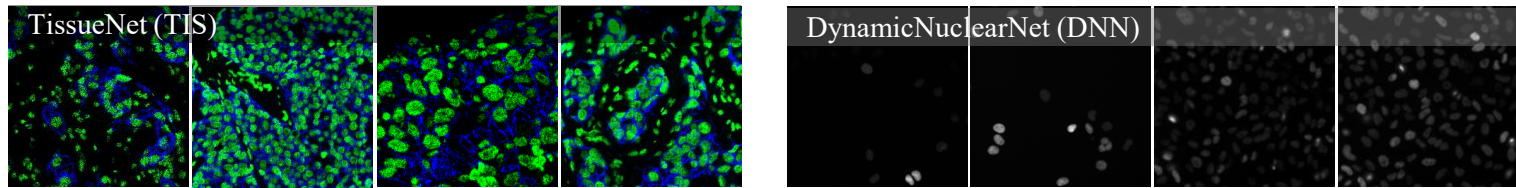
Software Download

Demo dataset download

Download the ApuShape demo dataset from

Google Drive: https://drive.google.com/drive/folders/1QRJIlvTRgpqTogpsJG8QINZVRSeeVRU7?usp=drive_link

Baidu Netdisk: <https://pan.baidu.com/s/1w5gjl0VRqmStGU1tRIVZrw?pwd=1111>



Images/

Two subfolders are provided: TIS and DNN. Each contains example images: TIS from TissueNet (fixed 256×256 px) ^[1] and DNN from DynamicNuclearNet (fixed 512×512 px) ^[2].

Models/

Two pretrained ShapeStar checkpoints are included: **tis.pth** (trained on TissueNet) and **dnn.pth** (trained on DynamicNuclearNet). Training details and evaluation metrics are provided in our paper.

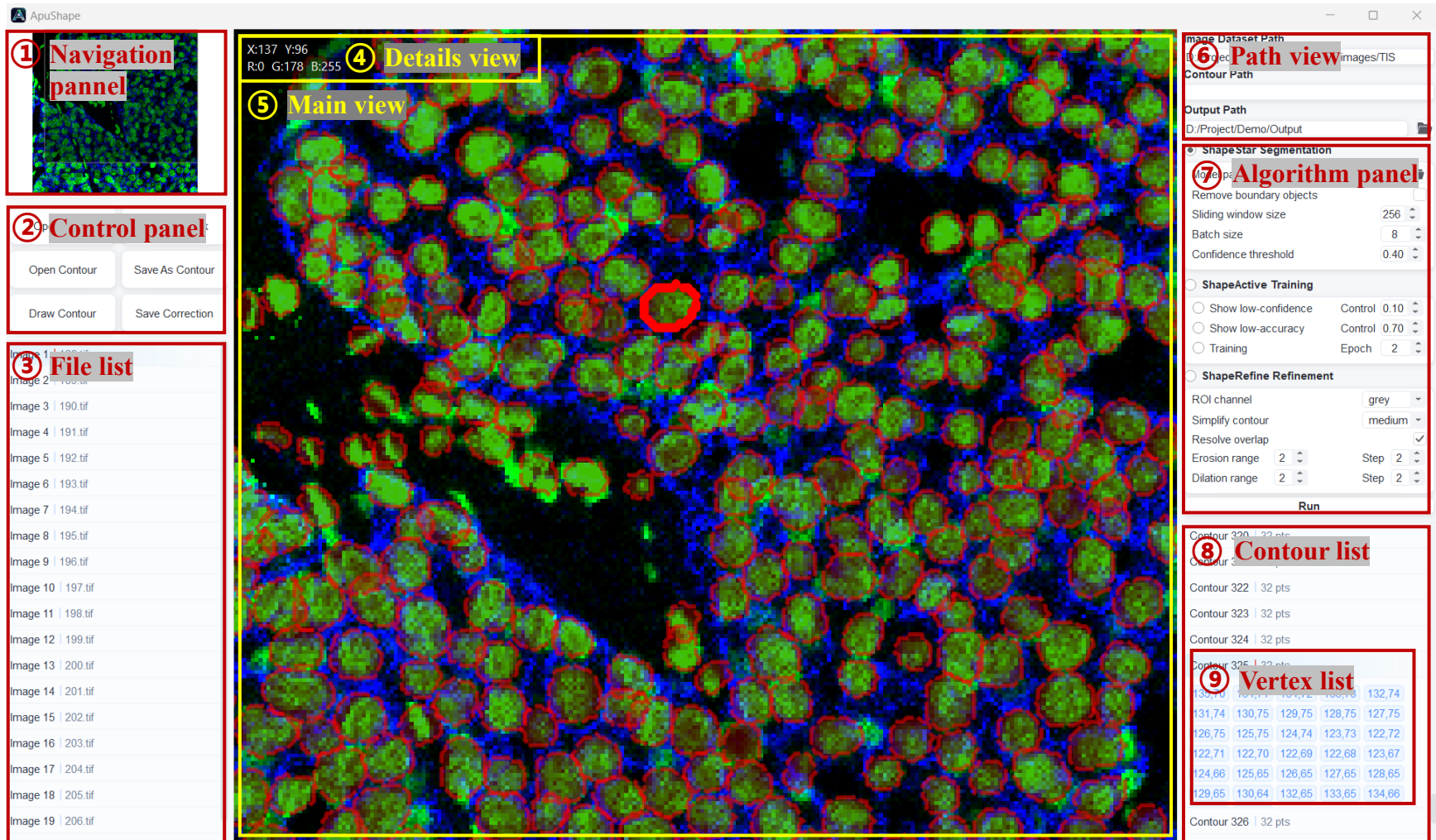
External/

Reference segmentation masks predicted by third-party tools: **Cellpose**, **CellSeg**, **HoVer-Net**, **Mesmer** and **Stardist**. These can be imported into ApuShape for review, correction and refinement.

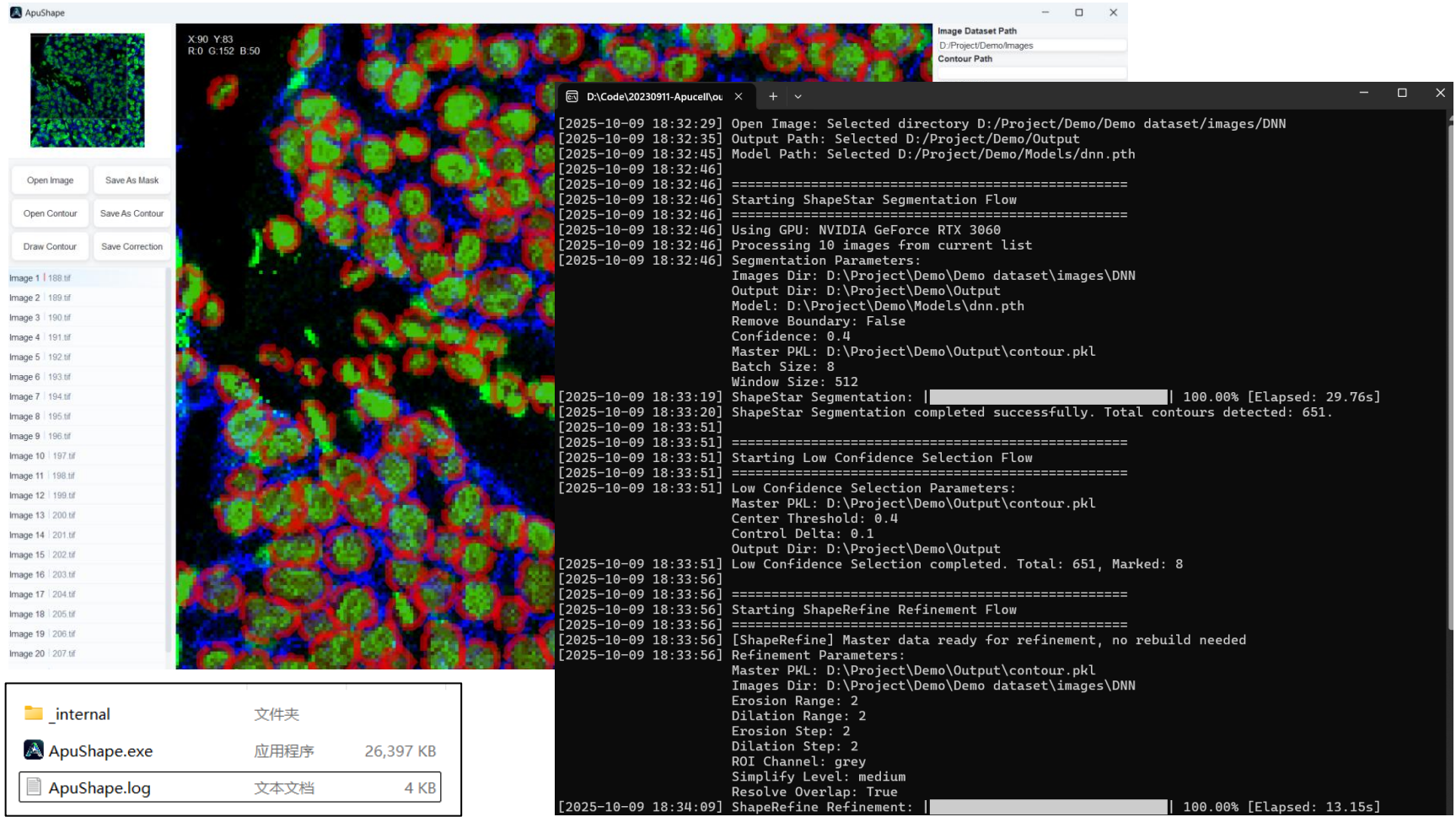
[1] <https://deepcell.readthedocs.io/en/master/data-gallery/tissuenet.html>

[2] <https://deepcell.readthedocs.io/en/latest/data-gallery/dynamicnuclearnet.html>

Main interface of ApuShape



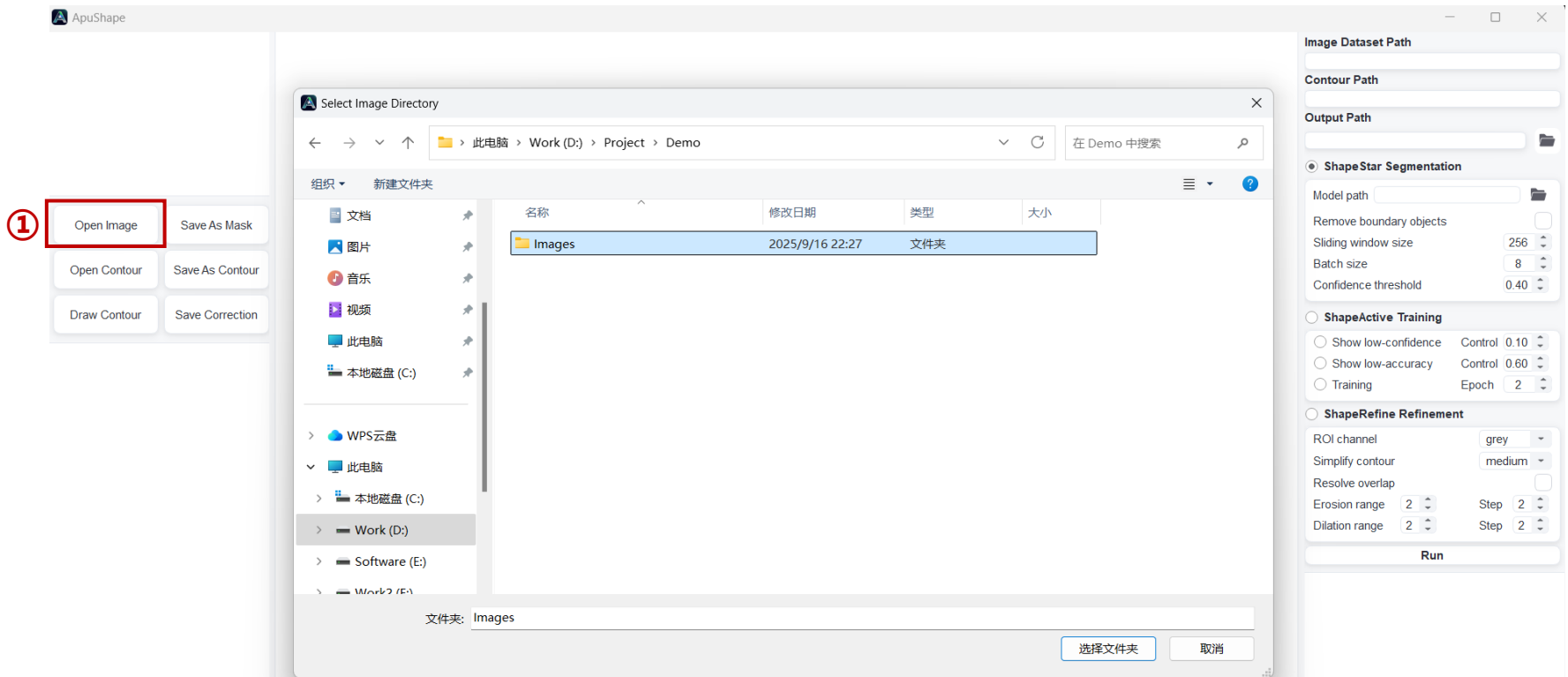
Main interface of ApuShape



The application streams logs to the **Console window** and also writes them to **ApuShape.log**.₆

Manual annotate datasets

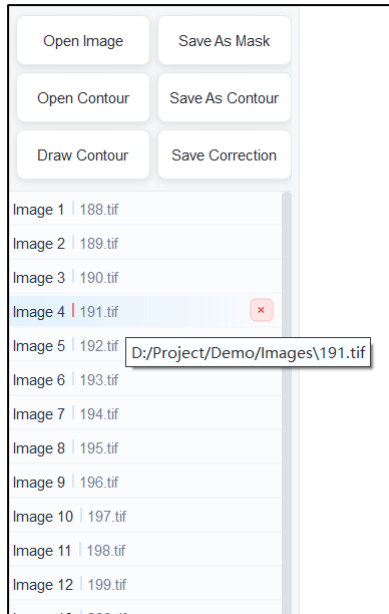
1. Select input images dictionary



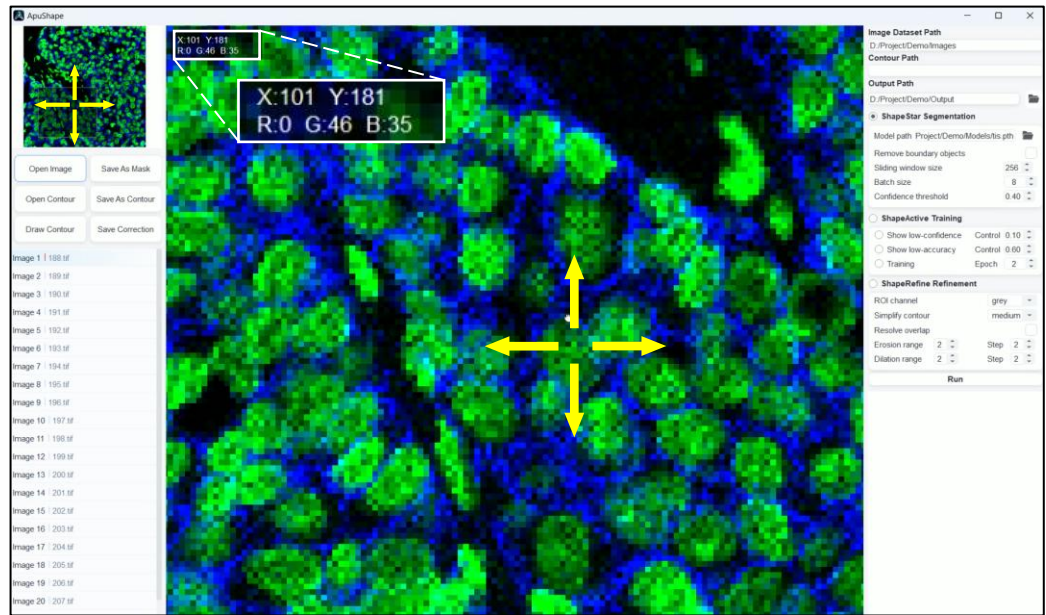
Supports JPEG (.jpg/.jpeg), PNG (.png), BMP (.bmp), and TIFF (.tif/.tiff) input images.

Manual annotate datasets

2. Browse the file list, zoom and pan to view images



Click an image to view it, **hover** to see the path, or click (✕) to **remove** it (**undoable** with Ctrl+Z).



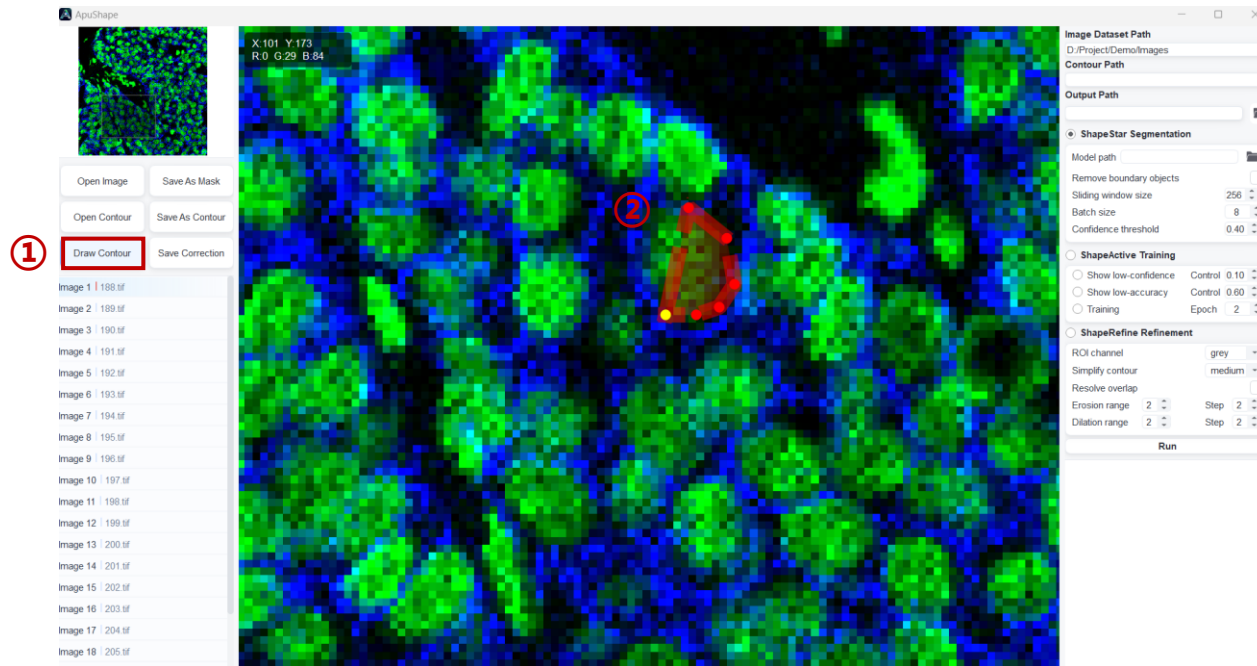
Hover in the Main View to see pixel coordinates and RGB values.

Use the **scroll wheel** to **zoom** the image, centered on your cursor.

Pan the view by **clicking** and **dragging** the image, or by **moving** the selection box in the Navigation panel.

Manual annotate datasets

3. Annotate with the contour tool



- ① Select the **Draw Contour** tool in the **Control Panel** (Shortcut: **D**; press **ESC** to cancel tool selection).
- ② **Left-click** along the object's edge to add points; **right-click** to undo the last point; the active point is shown in **yellow**. **Double-left-click** to finish the contour. **Double-right-click** cancels the current contour but keeps the tool active.

Manual annotate datasets

4. Select and remove contours

The screenshot shows the ApuShape software interface. The main view displays a segmented image with green and red regions. A yellow arrow points from a contour in the main view to its entry in the contour list. The contour list shows 'Contour 109 | 32 pts' with a red 'X' button next to it. A red circle highlights the 'X' button.

Contour 109 | 32 pts

224,147	223,148	223,149	219,150
222,151	221,152	220,152	219,153
218,153	217,153	216,153	214,153
213,152	213,151	212,149	212,148
213,147	213,146	213,145	214,144
214,143	215,143	216,143	217,143
218,142	219,143	220,143	221,143
222,143	222,144	223,145	223,146

Contour 110 | 32 pts

Contour 111 | 32 pts

Contour 112 | 32 pts

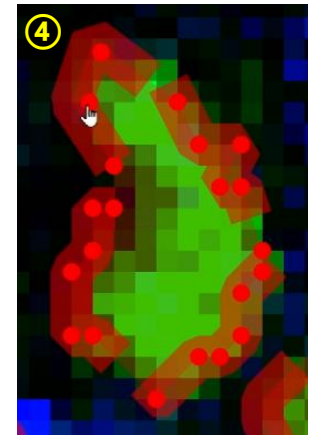
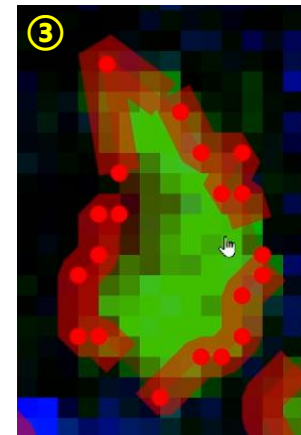
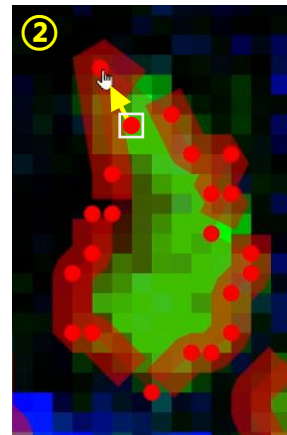
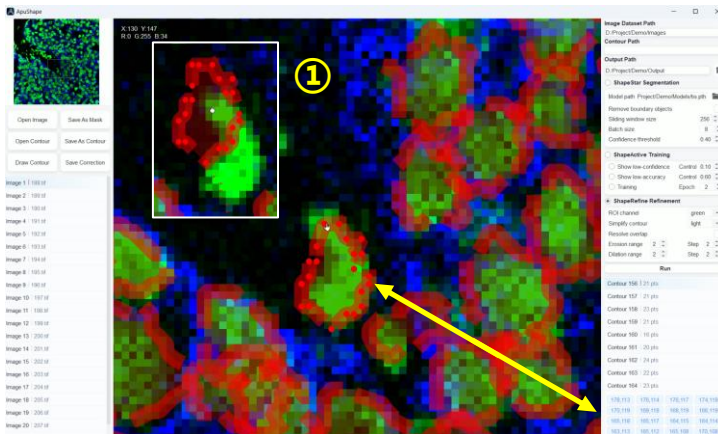
Contour 113 | 32 pts

Contour 114 | 32 pts

- ① Selections are synced between the **Main View** and the **Contour List**.
- ② **Double-click** a contour in the list to expand/collapse its **Vertex List**.
- ③ **Delete/Backspace** to remove a selected contour, or by clicking the (X) in the list (undoable with **Ctrl+Z**).

Manual annotate datasets

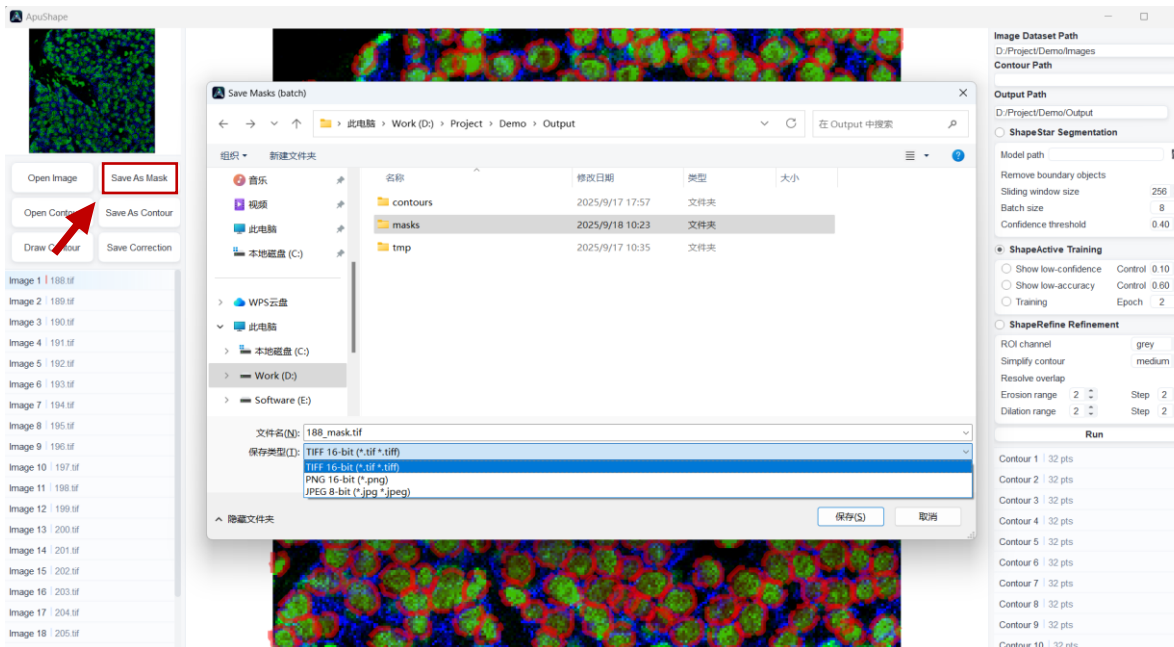
5. Correcting contour annotations



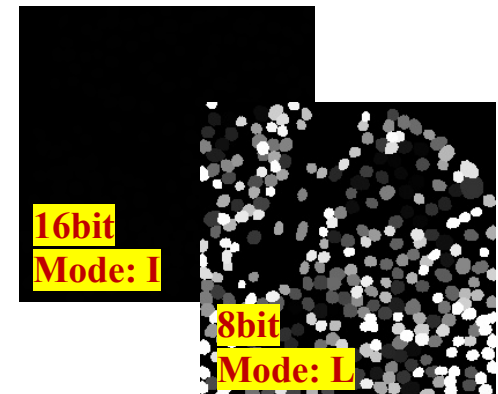
- ① **Click and drag** to move a contour (undoable with **Ctrl+Z**).
- ② **Click and drag** to move a vertex; **Vertex list** update in real time (undoable with **Ctrl+Z**).
- ③ **Right-double-click** a vertex or **press Delete/Backspace** to remove it (undoable with **Ctrl+Z**).
- ④ **Left-double-click** on the contour line to add a new vertex.

Manual annotate datasets

6. Export annotations as masks



192.tif	ImageGlass TIF F...	10 KB
193.tif	ImageGlass TIF F...	10 KB
194.tif	ImageGlass TIF F...	10 KB
195.tif	ImageGlass TIF F...	11 KB
196.tif	ImageGlass TIF F...	12 KB
197.tif	ImageGlass TIF F...	11 KB



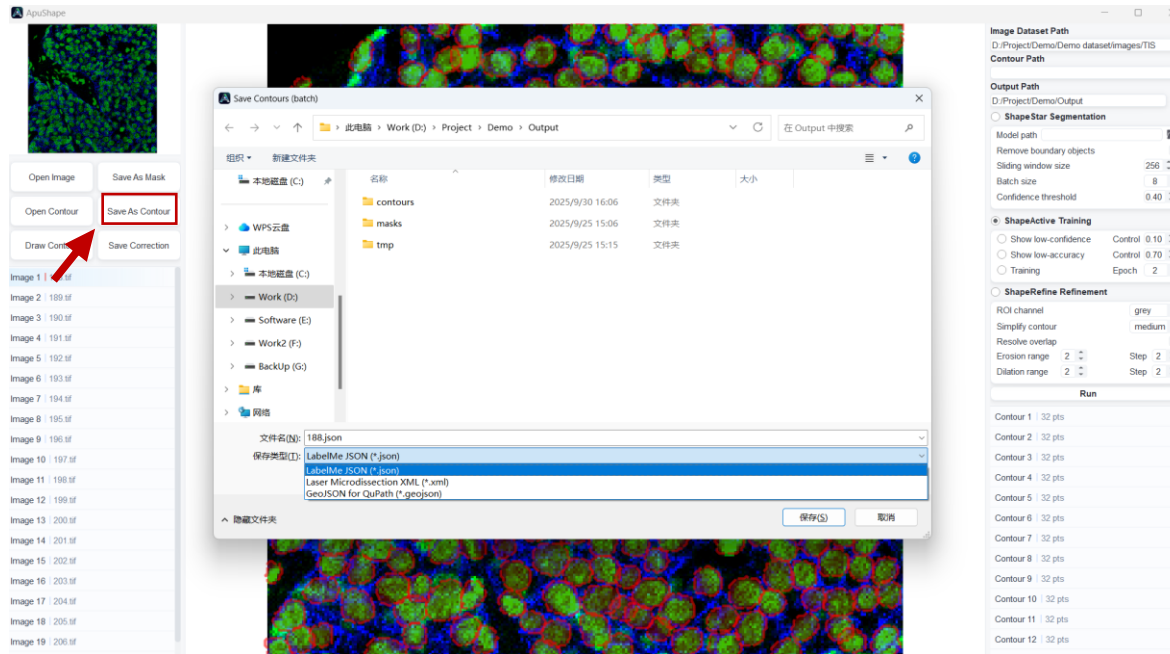
Choose an export format (TIFF/PNG/JPEG).

Note: Use **TIFF or PNG for 16-bit**, lossless masks. Use **JPEG for 8-bit**, compressed masks. JPEG can only store labels 1–255; if a mask has more than 256 objects, the extra ones are merged into one label (255).

Select an output folder.

Manual annotate datasets

7. Export annotations as contours



192.json	JSON File	288 KB
193.json	JSON File	303 KB
194.json	JSON File	297 KB
195.json	JSON File	288 KB
196.json	JSON File	373 KB
197.json	JSON File	328 KB

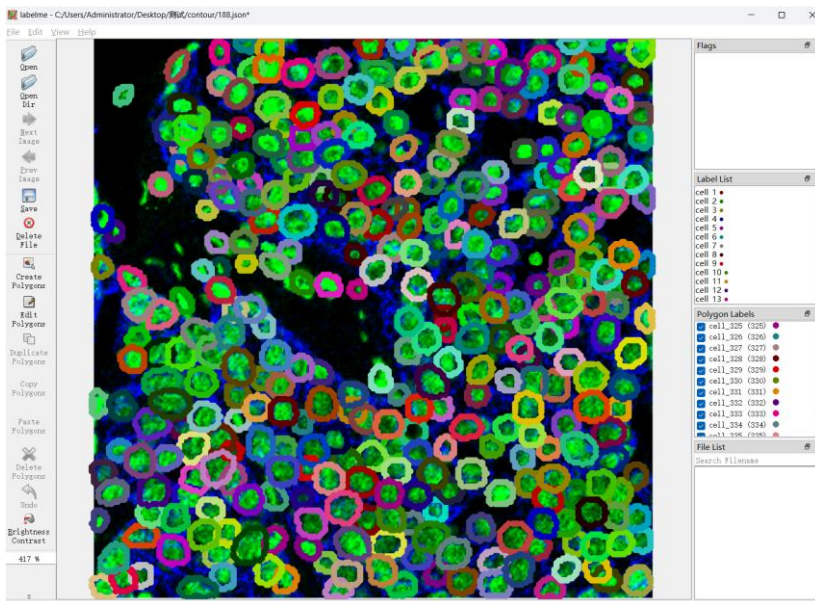
```
{  
  "type": "FeatureCollection",  
  "features": [  
    {  
      "type": "Feature",  
      "id": "1",  
      "geometry": {  
        "type": "Polygon",  
        "coordinates": [  
          [  
            [188.0,  
              15.0,  
              ],  
          ],  
          [187.0,  
            16.0,  
            ],  
          ],  
        ]  
      }  
    }  
  ]  
}
```

Choose an export format (LabelMe JSON, XML, GeoJSON for QuPath).

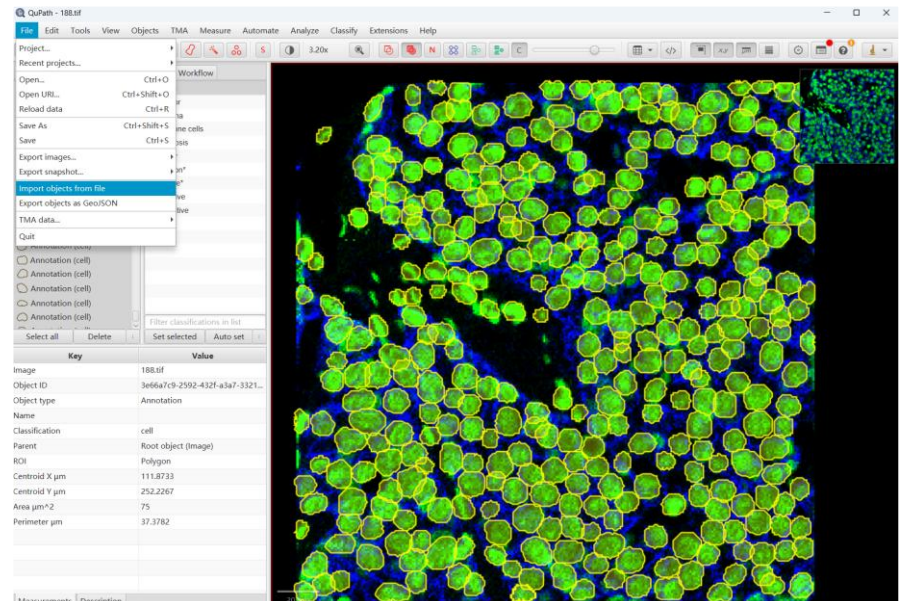
Select an output folder.

Manual annotate datasets

7. Export annotations as contours



LabelMe



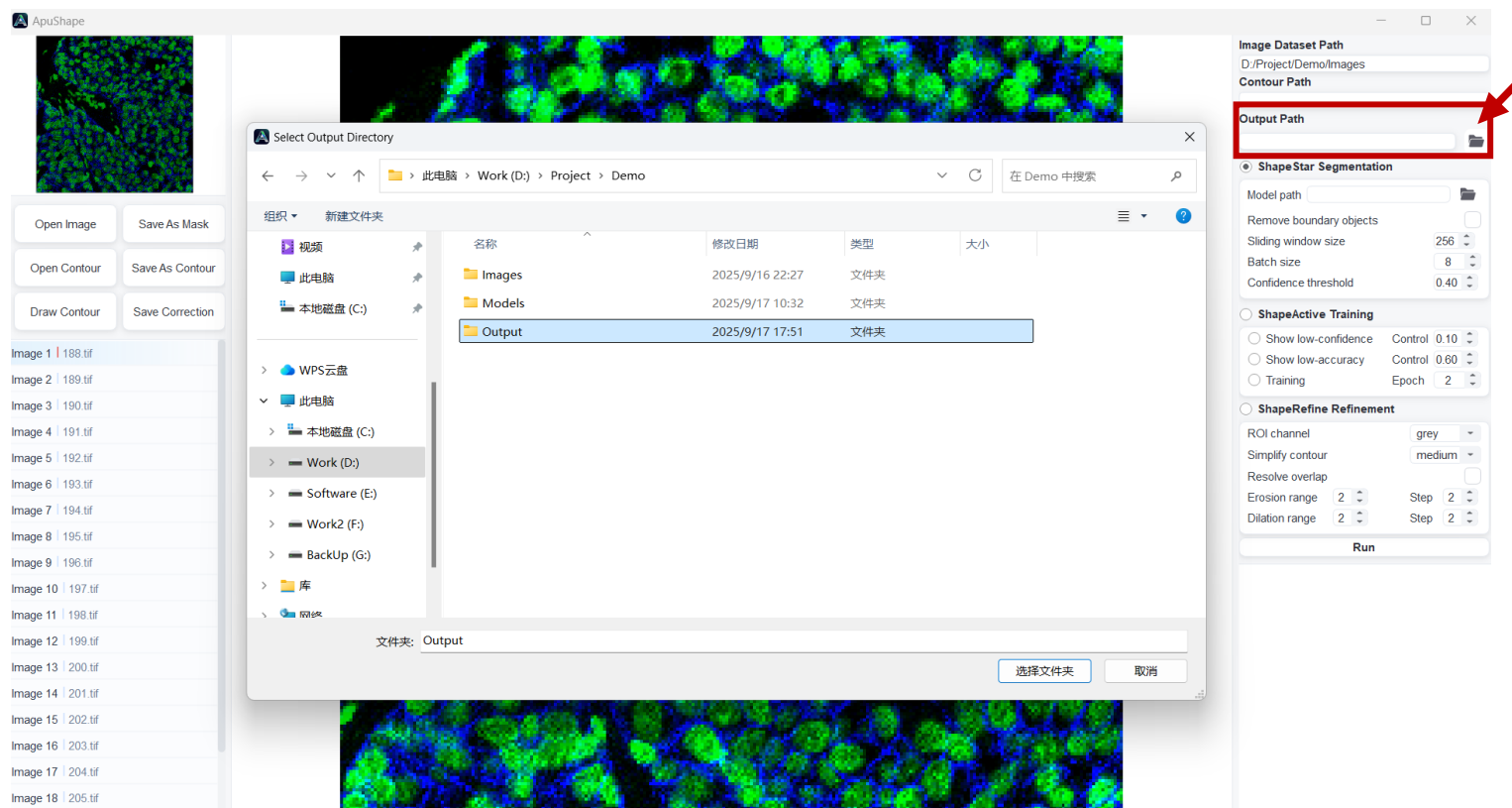
QuPath

The exported LabelMe JSON (.json) (using the version field **version: 4.6.0**) and GeoJSON for QuPath (.geojson) can be opened in LabelMe and QuPath, respectively.

Annotate datasets with ShapeStar

1. Select input images dictionary

2. Select output path



Annotate datasets with ShapeStar

3. Fill in the ShapeStar Segmentation section of the Algorithm panel.

3.1 Turn on the **radio button** next to ShapeStar Segmentation.

3.2 Select the **Model path** (details in page 27).

3.3 Check **Remove boundary objects**.

Note: When enabled, objects touching the image border are removed.

3.4 Set **Sliding window size**.

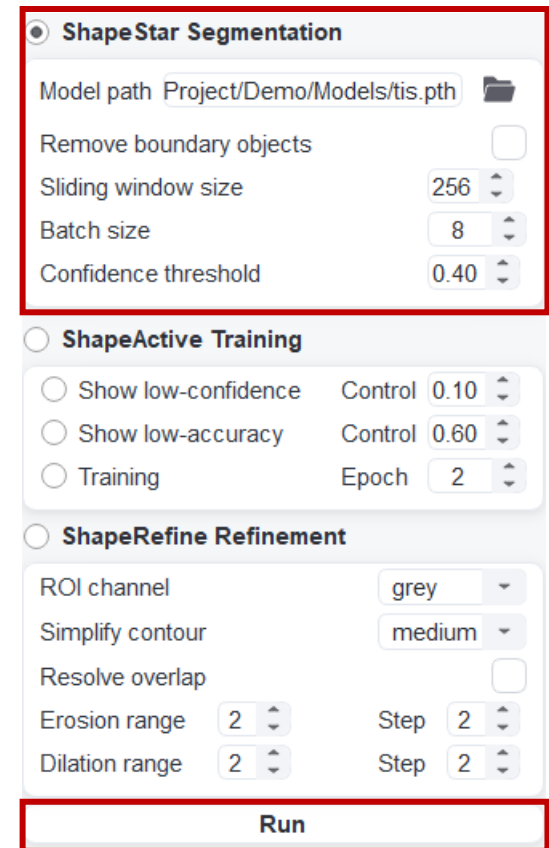
Note: Split large images into tiles with 25% overlap, then blend outputs in the overlapping regions.

3.5 Set **Batch size**

3.6 Set **Confidence threshold** ◇

Note: Keep only predictions with a score $\geq$ threshold. Raise it for fewer, more reliable contours.

3.7 Click **Run** to start.



The screenshot shows the 'ShapeStar Segmentation' panel with the following settings:

- ShapeStar Segmentation** (selected with a radio button)
- Model path: Project/Demo/Models/tis.pth
- Remove boundary objects: ☐
- Sliding window size: 256
- Batch size: 8
- Confidence threshold: 0.40

Below this panel are two other sections:

- ShapeActive Training** (radio button selected):
 - Show low-confidence: Control 0.10
 - Show low-accuracy: Control 0.60
 - Training: Epoch 2
- ShapeRefine Refinement** (radio button selected):
 - ROI channel: grey
 - Simplify contour: medium
 - Resolve overlap: ☐
 - Erosion range: 2, Step: 2
 - Dilation range: 2, Step: 2

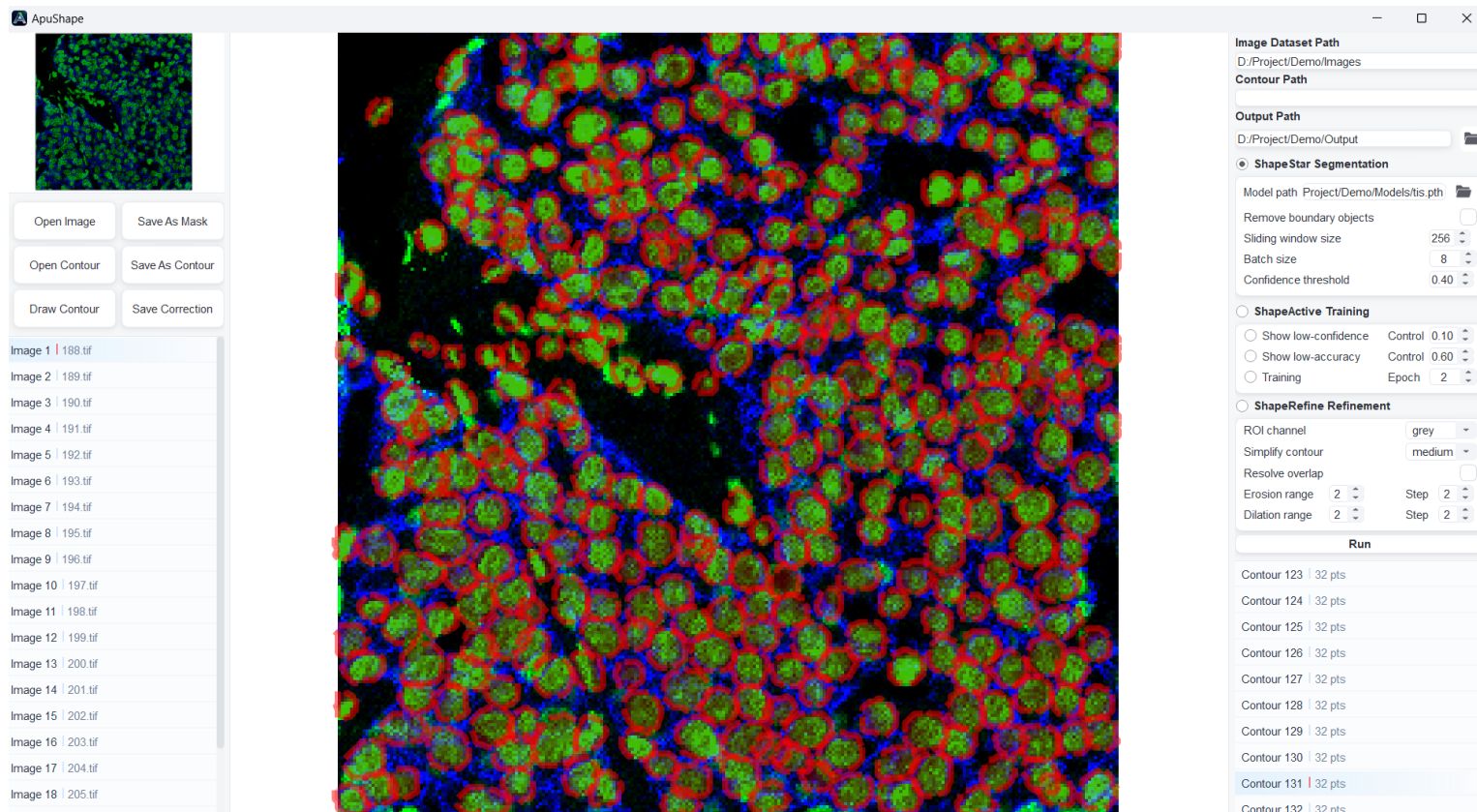
A large **Run** button is located at the bottom of the panel.

◇ With a 13th-Gen Intel Core i5-13600KF and an RTX 3060, **256×256 TissueNet images** run in **~1 s/image**.

◇ **Recommended:** Confidence threshold ≈ 0.4 for TIS dataset, ≈ 0.75 for DNN dataset.

Annotate datasets with ShapeStar

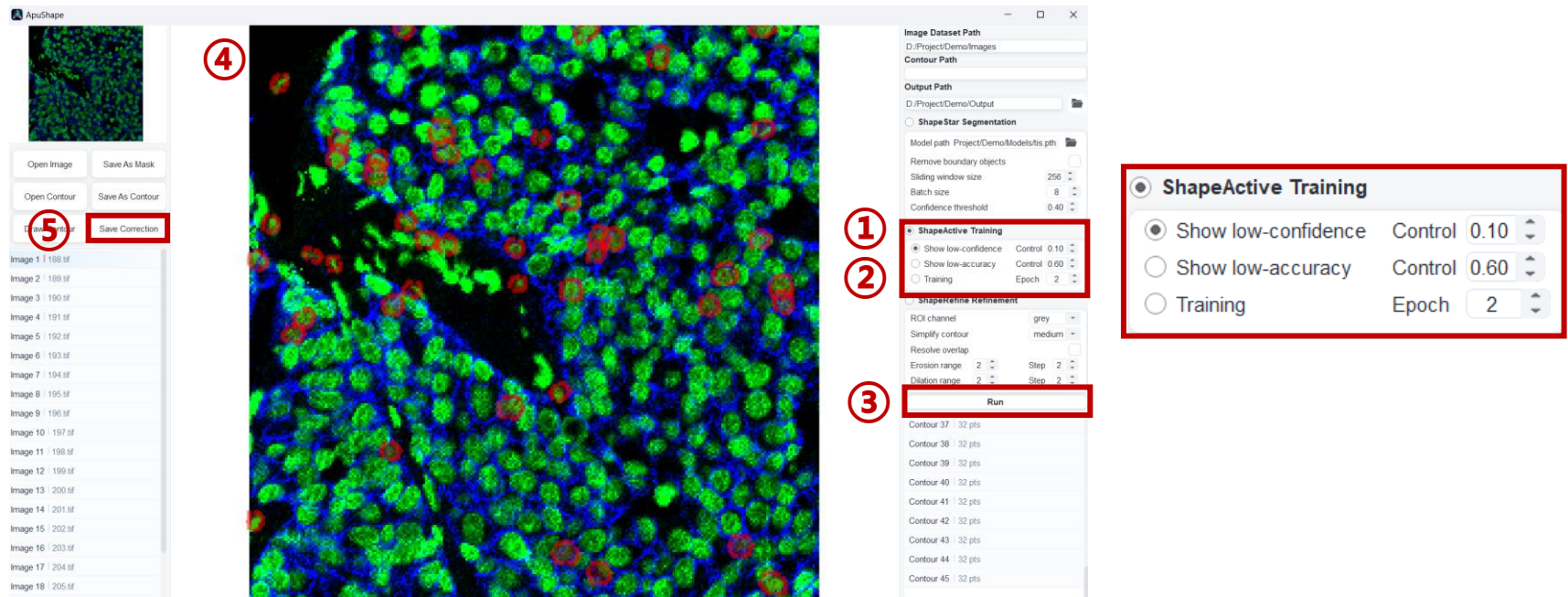
4. View, correct and save ShapeStar Segmentation results



Each annotation is a **32-vertex** polygon.

Correct annotations / retrain model

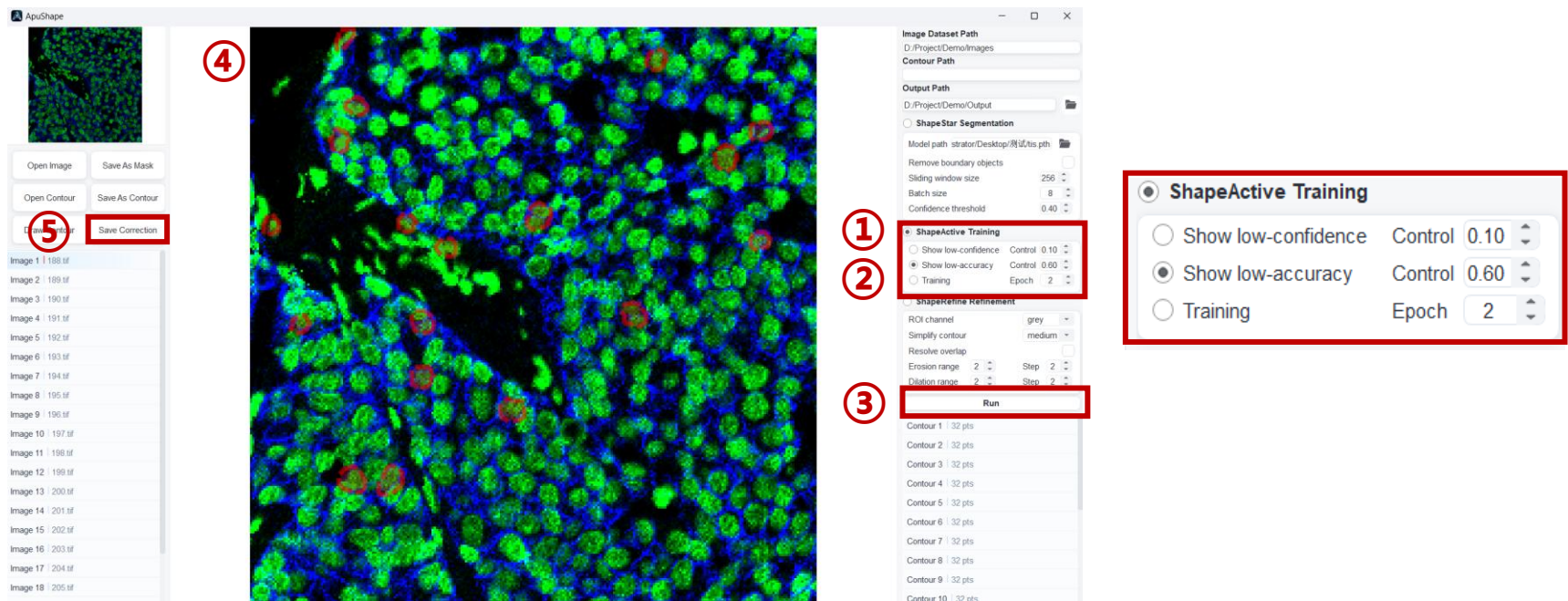
1. Show low-confidence objects



- ① Turn on the radio buttons next to **ShapeActive Training** and **Show low-confidence**.
- ② Set **Control**: instances within **[Confidence threshold $\pm$ Control]** are highlighted as low-confidence.
- ③ Click **Run** to start.
- ④ **Add/delete** and **correct** contours.
- ⑤ Click **Save correction** to save your changes; temporary contours are stored at **Output Path/contout.pkl**.

Correct annotations / retrain model

2. Show low-accuracy objects



① Turn on the radio buttons next to **ShapeActive Training** and **Show low-accuracy**.

② Set **Control**: instances near **[Control]** are highlighted as low-accuracy ◇.

③ Click **Run** to start.

④ **Add/delete** and **correct** contour vertices.

⑤ Click **Save correction** to save your changes; temporary contours are stored at **Output Path/contout.pkl**.

◇ **Recommended**: Low-accuracy control ≈ 0.75 for TIS dataset, ≈ 0.90 for DNN dataset.

Correct annotations / retrain model

3. Show current contours

The screenshot shows the ApuShape software interface. On the left is a file list with images 1 through 18. The main area displays a large image with green and red contours. On the right is a settings panel. The 'ShapeActive Training' section has three radio buttons: 'Show low-confidence', 'Show low-accuracy', and 'Training'. The 'Training' radio button is selected. The 'ShapeActive Refinement' section has a 'Run' button. The 'Run' button is highlighted with a red box and a circled '2'.

Image Dataset Path: D:/Project/Demo/Images
Contour Path:

ShapeActive Training

- ☐ Show low-confidence Control 0.10
- ☐ Show low-accuracy Control 0.60
- ☒ Training Epoch 2

ShapeActive Refinement

ROI channel: grey
Simplify contour: medium
Resolve overlap: ☐
Erosion range: 2 Step: 2
Dilation range: 2 Step: 2

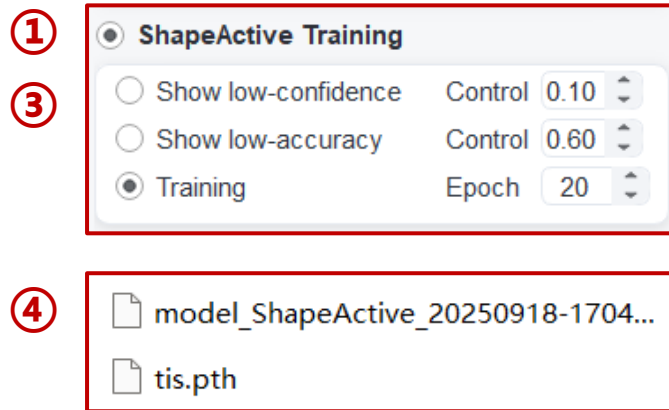
Run

Contour 37 | 32 pts
Contour 38 | 32 pts
Contour 39 | 32 pts
Contour 40 | 32 pts
Contour 41 | 32 pts
Contour 42 | 32 pts
Contour 43 | 32 pts
Contour 44 | 32 pts
Contour 45 | 32 pts
Contour 46 | 32 pts

- ① Turn on the radio button next to **ShapeActive Training**, and **deselect** all sub-options below.
- ② Click **Run** to display the current contours from **Output Path/contout.pkl**.

Correct annotations / retrain model

4. Retrain the Model with Corrections



```
Starting ShapeActive Training Flow
Continue training based on model: D:\Project\Demo\Models\dnn.pth Auto-saving corrections before
No changes to save
Training Parameters:
Master PKL: D:\Project\Demo\Output\contour.pkl
Epochs: 20
Batch Size: 4
Old Model: D:\Project\Demo\Models\dnn.pth
Val Ratio: 0.2
High Weight: 5.0
Training Samples: 450
Data split: 8 train, 2 val
Weight training: 0 corrected, 0 removed, 0 added, 450 total contours
Using GPU: NVIDIA GeForce RTX 3060
Baseline AJI: 0.8789
Epoch 1/20: 100.00% [Elapsed: 2.79s]
Loss: 0.4236 (Prob: 0.1042, Dist: 31.9478)
Validation AJI: 0.8330
New best AJI: 0.8330
Epoch 2/20: 100.00% [Elapsed: 2.36s]
Loss: 0.4335 (Prob: 0.1068, Dist: 32.6716)
Validation AJI: 0.8770
New best AJI: 0.8770
```

- ① Turn on the radio buttons next to ShapeActive Training and Training.
- ② Set **Epoch** and Batch size $\diamond$.
- ③ Click **Run** to start **fine-tuning**; corrected/new objects are weighted higher. If **Model path** under ShapeStar Segmentation is **not** set, training starts from **scratch**.
- ④ New weights are saved beside the original model with the filename suffix:
“_ShapeActive<timestamp>.pth”

$\diamond$ Training batch size is set to half the segmentation batch size, configured under ShapeStar Segmentation.

Refine annotations

1. Fill in the ShapeRefine Refinement section of the Algorithm panel.

1.1 Turn on the radio button next to ShapeRefine Segmentation.

1.2 Select the **ROI channel**.

Note: Choose the staining channel that highlights the target cells/nuclei.

1.3 Choose **Contour simplification** strength.

Note: Reduce vertices while preserving shape (None/Light/Medium/Strong).

1.4 Check **Resolve overlap**.

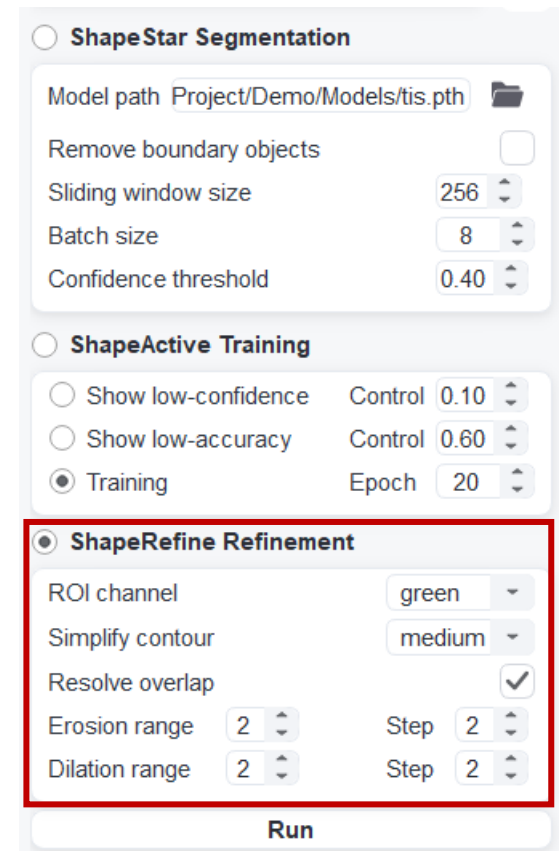
Note: Remove overlaps between instance contours.

1.5 Set **Erosion/Dilation** range and step.

Range: pixel sampling window around the contour for evaluating vertex updates; higher values mean stronger per-vertex update strength.

Step: number of vertex iteration; higher values allow greater overall boundary movement.

1.6 Click **Run** to start.



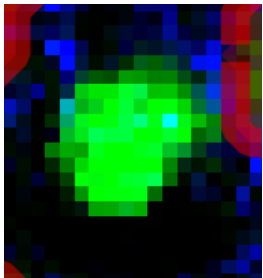
The screenshot shows the 'ShapeRefine Refinement' section of the Algorithm panel, which is highlighted with a red border. The section includes the following settings:

- ROI channel:** green (dropdown menu)
- Simplify contour:** medium (dropdown menu)
- Resolve overlap:** checked (checkbox)
- Erosion range:** 2 (spinner)
- Erosion Step:** 2 (spinner)
- Dilation range:** 2 (spinner)
- Dilation Step:** 2 (spinner)

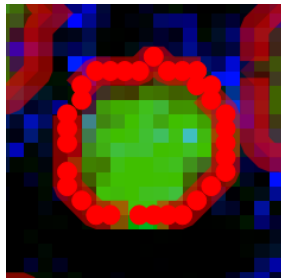
Below the settings is a 'Run' button.

Refine annotations

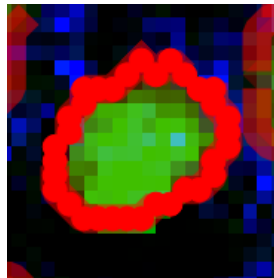
Simplification strength:



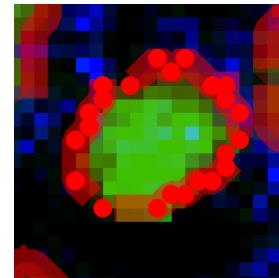
Input



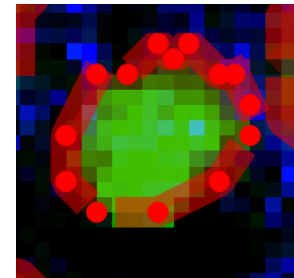
ShapeStar



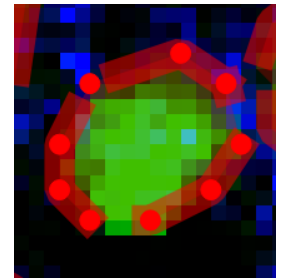
None



Light

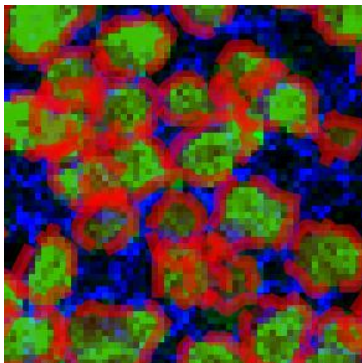


Medium

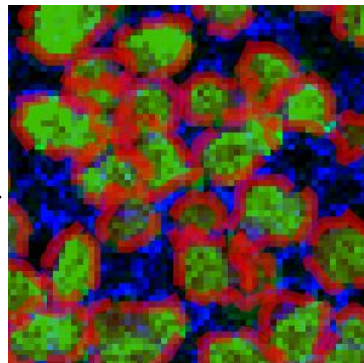
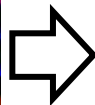


Strong

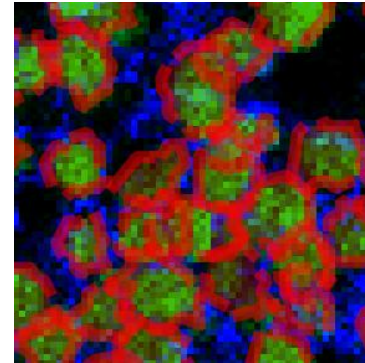
Resolve overlap:



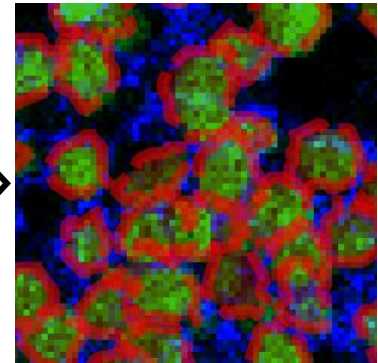
Off



On



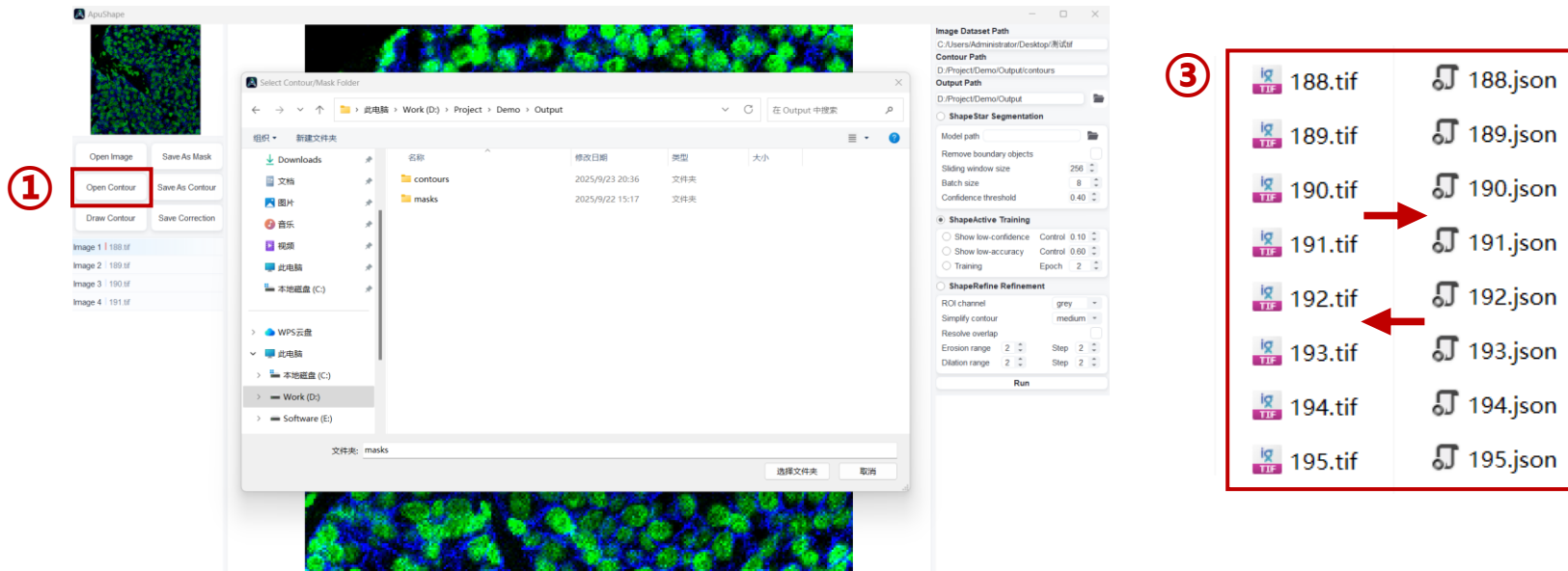
Off



On

Refine annotations

2. Import and refine external contours



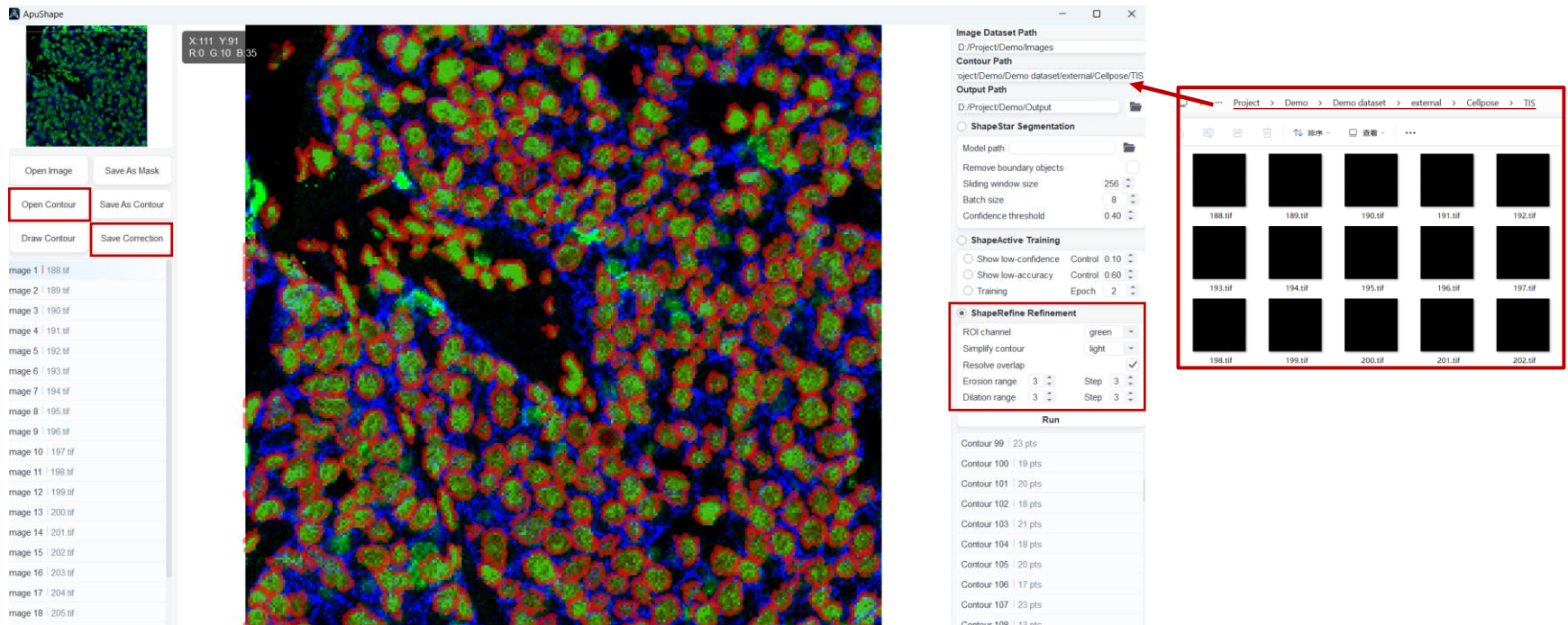
① Import **external** segmentation or annotation results (LabelMe JSON and GeoJSON) into ApuShape for review, correction, and refinement with ShapeRefine.

② Mask files (TIFF, PNG, JPEG) can also be imported and are **automatically converted** to labels, but this workflow is not recommended due to sampling artifacts.

③ For file pairing, keep **identical filenames** and differ only by extension.

Refine annotations

EX 1: Import Cellpose predictions. Fill in the ShapeRefine Refinement section of the Algorithm panel, then click Run to start.



Click **Save Correction** to save the refined contours, or use **Show Current Contours** to return to the pre-refinement contours; adjust parameters and run refinement again as needed.

Note

Display

- For optimal display, we recommend a screen resolution of **2560 × 1440** with **125%** scaling. If you encounter interface issues such as blurred text, misaligned layout, or offset mouse clicks, please adjust your system's resolution and scaling as suggested, or just lower the scaling.

GPU acceleration

- ApuShape now supports GPU acceleration for faster processing.

Requirements: NVIDIA GPU (GTX 1050 or newer) and the NVIDIA driver installed.

To verify the driver, run `nvidia-smi` in the Console window.

Task output / resuming

- For every task, **output.pkl** is essential. Before starting a new task, select a clean (empty) output folder. To continue a previous task, select the same output folder so ApuShape can reuse the existing `output.pkl`.

Note

Model Selection

- Two pretrained models are provided: `dnn.pth` (trained on **single-channel** stained DNN data) and `tis.pth` (trained on **dual-channel** stained TIS data). To use `tis.pth`, convert the ROI channel to the green and the reference channel to blue. To use `dnn.pth`, simply convert images to grayscale.

Simplify Contour

- The **Resolve Overlap** algorithm automatically resamples contour points, which can increase their count.

If **Resolve Overlap** is enabled without **Simplify Contour**, the system will automatically simplify each contour to approximately 32 points.